



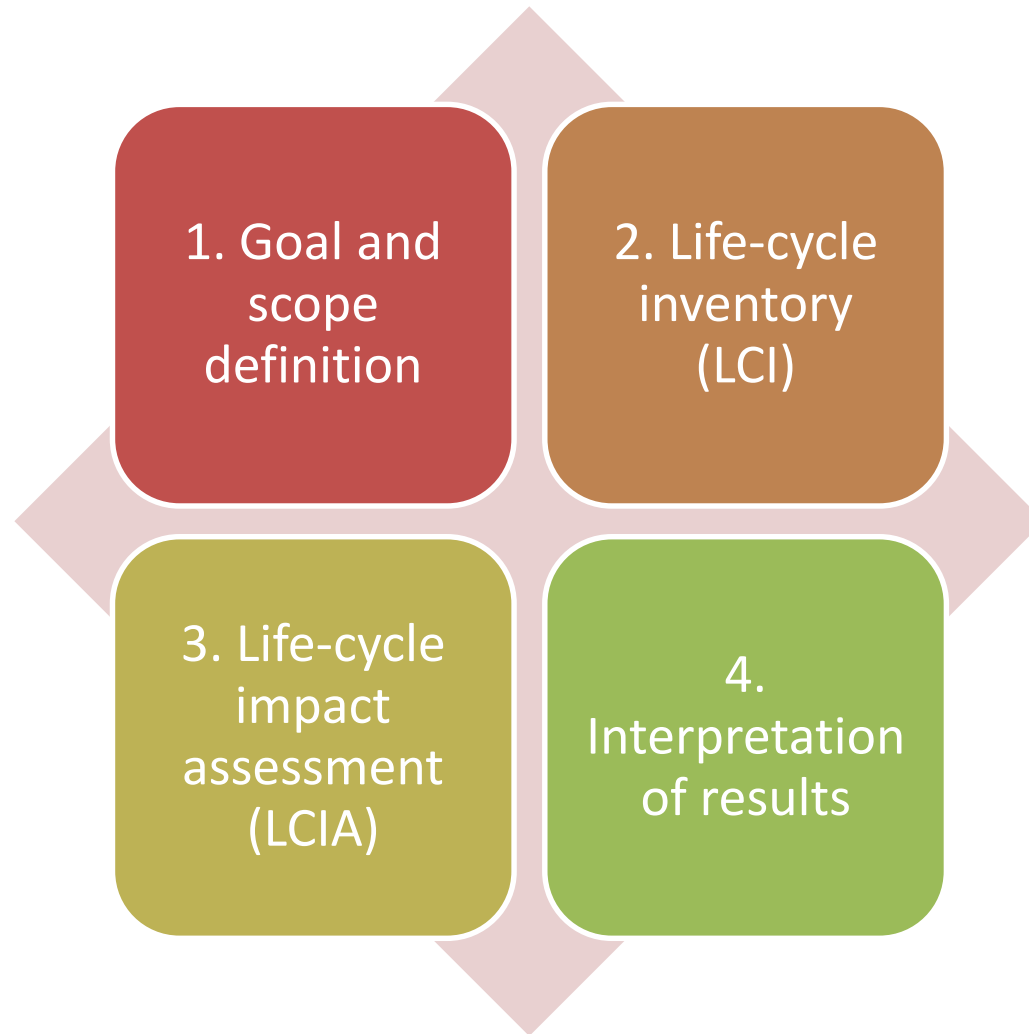
Sustainable Engineering 2: Life cycle analysis

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Materials life cycle analysis (LCA)

- A life-cycle assessment (LCA) is a tool that can be used to evaluate the potential environmental impacts of a product, material, process, or activity.
- An LCA is a comprehensive method for assessing a range of environmental impacts across the full life cycle of a product system, from materials acquisition to manufacturing, use, and final disposition.

LCA process (ISO 14040 – 2006)



LCA process (ISO – 2006)

Life cycle of a material can be represented using an input/output diagram

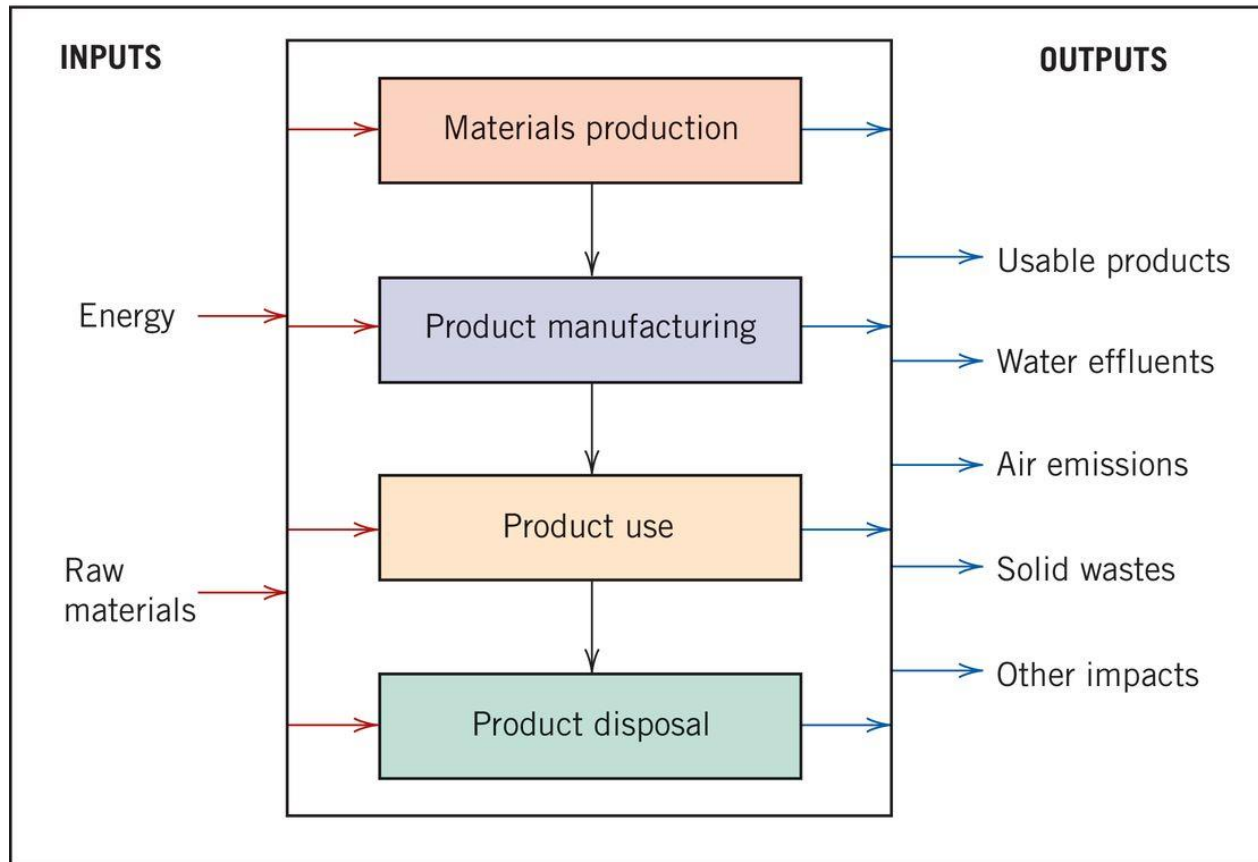


Fig. 23.2, Callister & Rethwisch 9e..

Input – Output Analysis

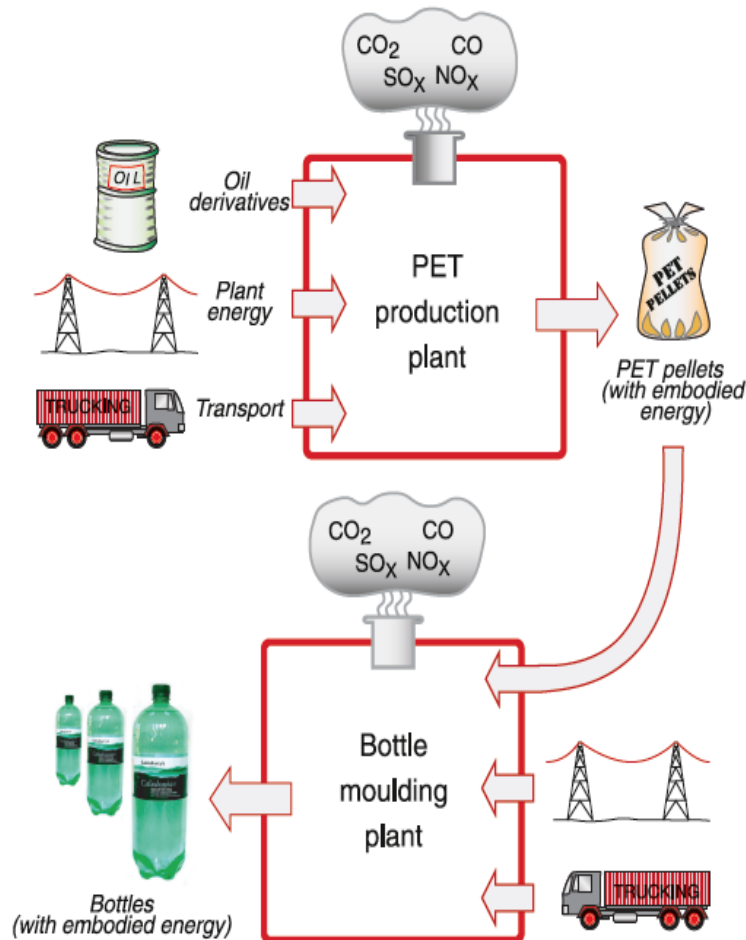
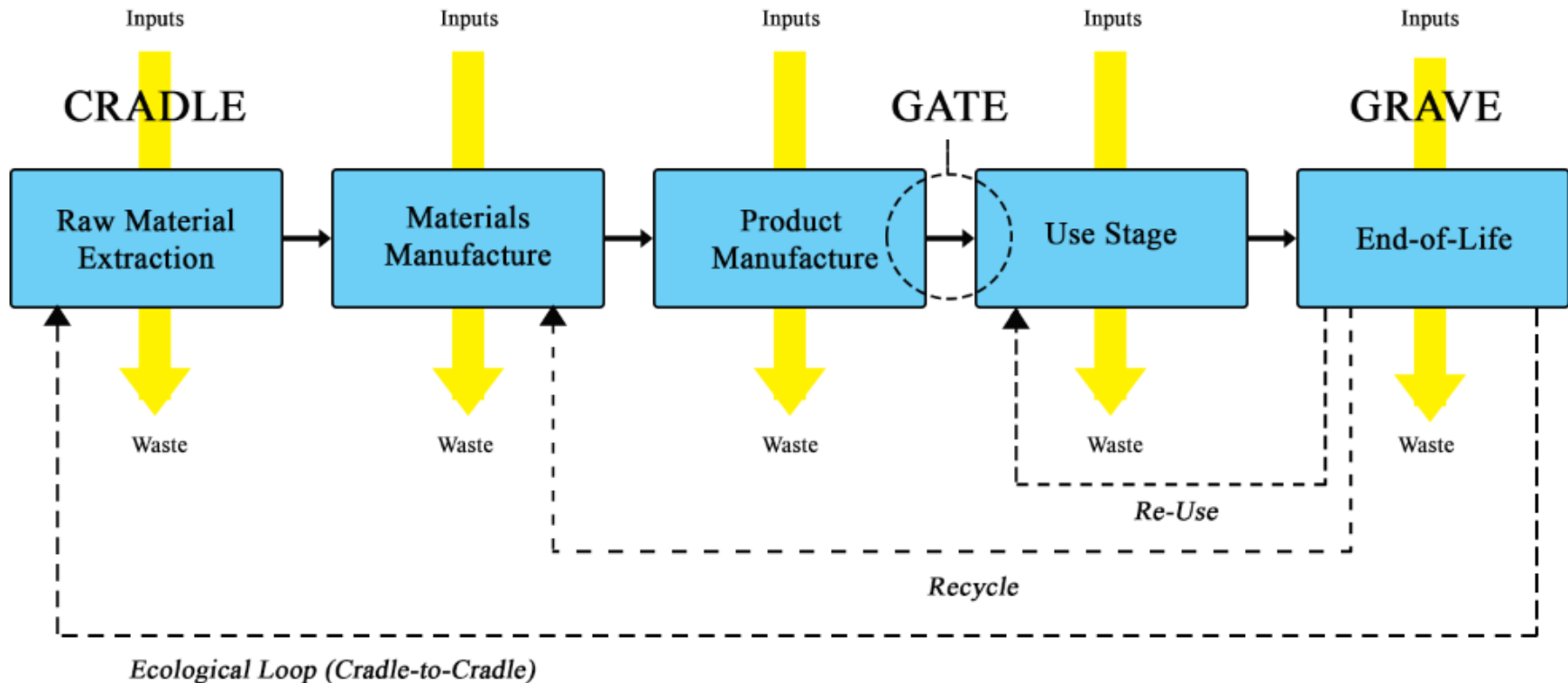


Figure 20.4

- Used to assess embodied energies
- The processing energy H_p is the energy used to shape, join, and finish 1 kg of the material to create a component or product
- Inputs to this analysis involve the processing energy and the material's embodied energy
- The output is the energy committed per part produced

Life-Cycle Assessment and Energy

Cradle-to-Grave cycle



Cradle-to-Cradle cycle

Life-Cycle Assessment and Energy

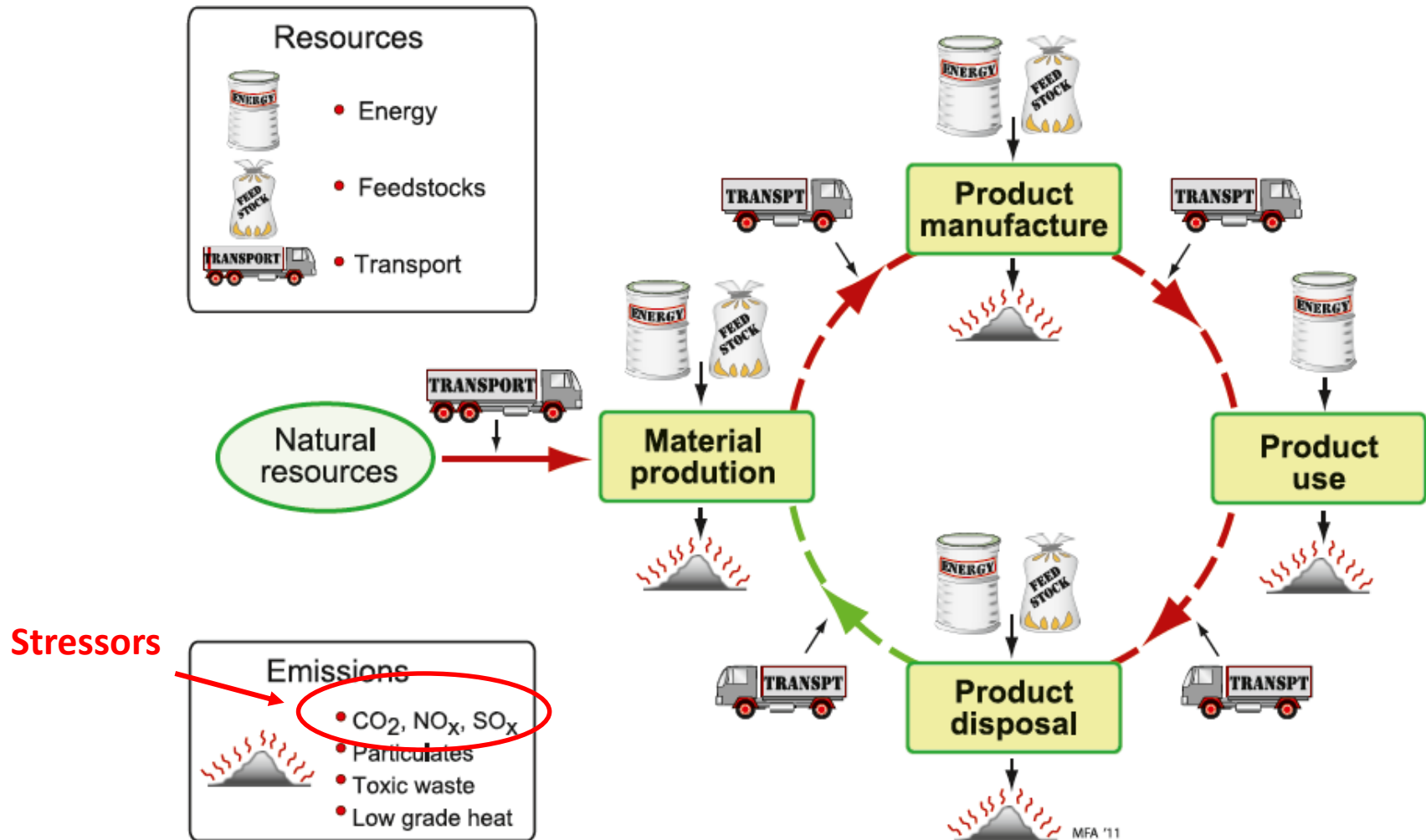


Figure 20.3, Ashby

Definitions



- **The embodied energy H_m** (MJ/kg) of a material is the energy that must be committed to create 1 kg of usable material
- **The CO_2 footprint** is the associated release of CO_2 in kg/kg
- **The processing energy H_p** associated with a material is the energy, in MJ, used to shape, join and finish 1 kg of the material to create a component or product.
- **The end-of-life potential** summarises the possible utility of the material at life's end

Components of 'Green Design'

- **Reduce** – redesign the product to use less material

example: PET bottles with thinner walls



Christopher Steer/iStockphoto

- **Reuse** – fabricate the product of a material that can be reused
example: refillable bottles and shipping containers
example: grind up old tyres for use as soft surfaces
- **Recycle** – reprocess the material into a new product
example: convert PET bottles to carpet fibers

Recycling: Ideals and Realities

Table 23.1 Recycle Codes, Uses of Virgin Material, and Recycled Products for Several Commercial Polymers

<i>Recycle Code</i>	<i>Polymer Name</i>	<i>Uses of Virgin Material</i>	<i>Recycled Products</i>
	Poly(ethylene terephthalate) (PET or PETE)	Plastic beverage containers; mouthwash jars; peanut butter and salad dressing bottles	Liquid-soap bottles, strapping, fiberfill for winter coats, surfboards, paint brushes, fuzz on tennis balls, soft-drink bottles, film, egg cartons, skis, carpets, boats
	High-density polyethylene (HDPE)	Milk, water, and juice containers; grocery bags; toys; liquid detergent bottles	Soft-drink bottle base caps, flower pots, drain pipes, signs, stadium seats, trash cans, recycling bins, traffic-barrier cones, golf-bag liners, detergent bottles, toys
	Poly(vinyl chloride) or vinyl (V)	Clear food packaging; shampoo bottles	Floor mats, pipes, hoses, mud flaps
	Low-density polyethylene (LDPE)	Bread bags; frozen-food bags; grocery bags	Garbage can liners, grocery bags, multipurpose bags
	Polypropylene (PP)	Ketchup bottles; yogurt containers; margarine tubs, medicine bottles, carpet fibers	Manhole steps, paint buckets, ice scrapers, fast-food trays, lawnmower wheels, automobile battery parts
	Polystyrene (PS)	Compact disc jackets; coffee cups; knives, spoons, and forks; cafeteria trays; grocery store meat trays; fast-food sandwich containers	License-plate holders, golf-course and septic-tank drainage systems, desktop accessories, hanging files, food-service trays, flowerpots, trash cans

Source: Reproduced with permission from SPI: The Plastics Industry Trade Association.

Recycling: Ideals and Realities

End of life potential of a product

- Ability to be recycled into the product from which it came
- Ability to be down-cycled into a lower-grade application
- Ability to be biodegraded into usable compost
- Ability to yield energy by controlled combustion
- Ability to be buried as landfill

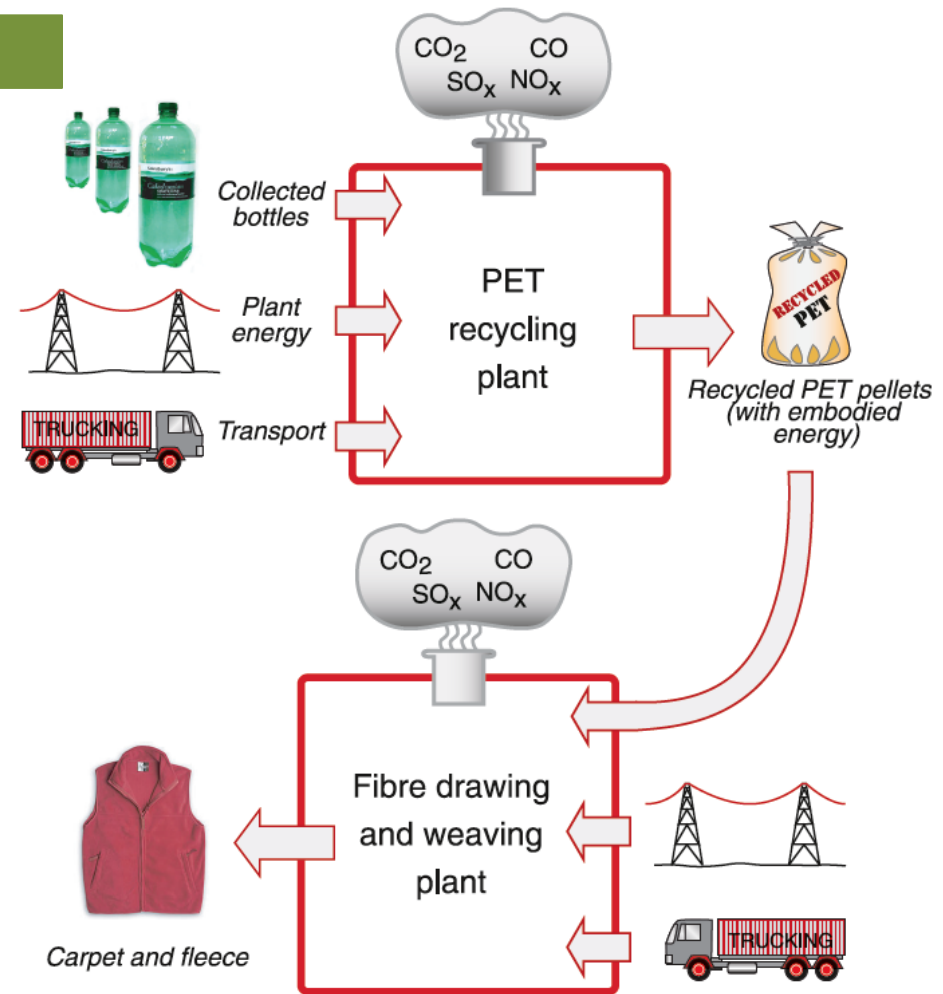


Figure 20.5, Ashby

Recycling: Ideals and Realities

Steps required to recycle PET, each on consuming energy

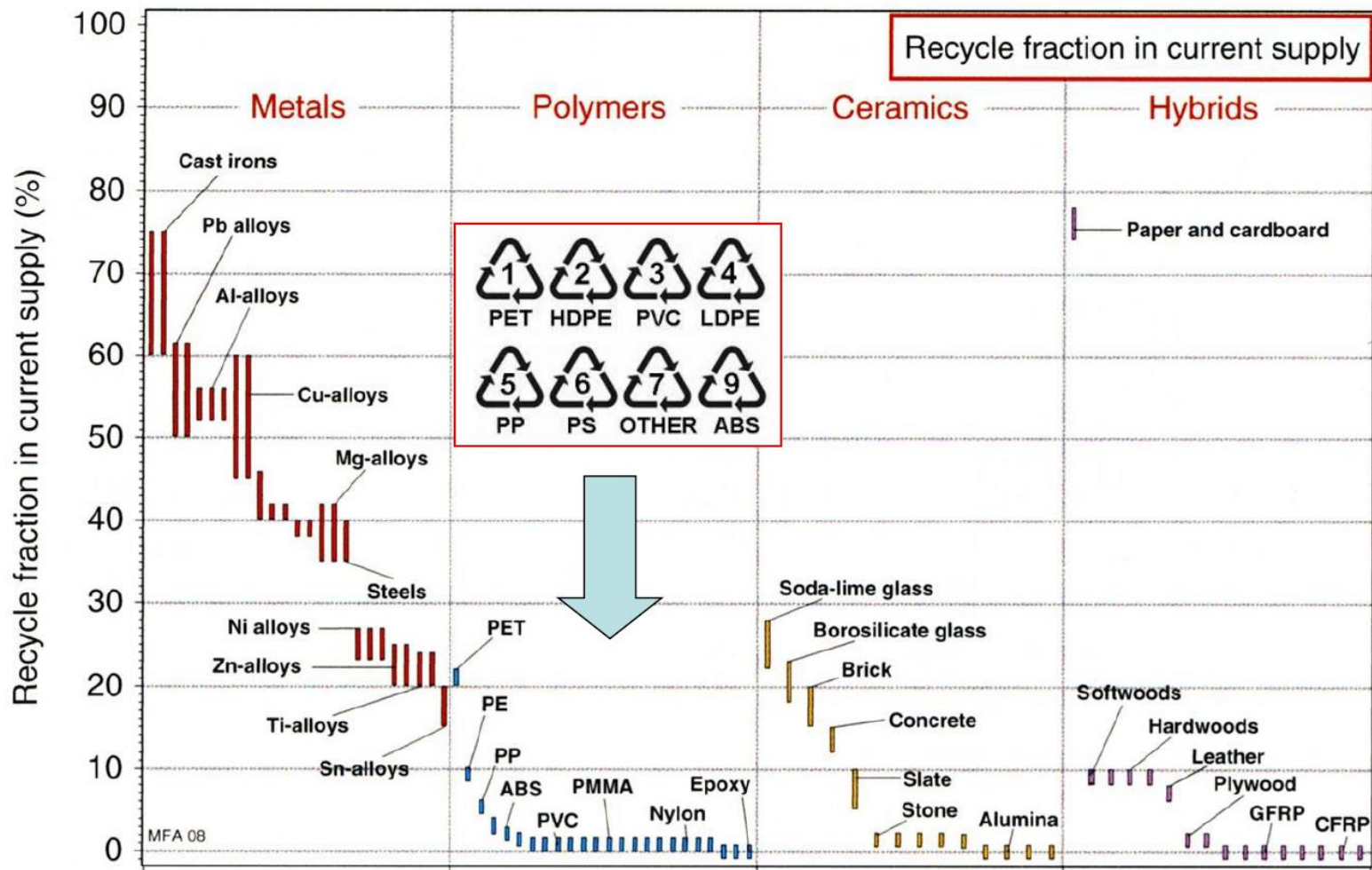
- | | |
|-------------------------|-----------------------------|
| 1. Collection | 7. Melting |
| 2. Inspection | 8. Filtration |
| 3. Chopping | 9. Pelletising |
| 4. Washing | 10. Packaging |
| 5. Flotation–separation | 11. Plant heating, lighting |
| 6. Drying | 12. Transport |

For polymers such as PET, about 50% of the embodied energy can be saved from recycling – this number can be much higher for certain metals but can also fall to zero for other materials

Polymer	Embodied energy* (MJ/kg)		Price [†] (\$/kg)	
	<i>Virgin</i>	<i>Recycled</i>	<i>Virgin</i>	<i>Recycled</i>
HDPE	82	40	1.9	0.9
PP	82	40	1.8	1.0
PET	85	55	2.0	1.1
PS	101	45	1.5	0.8
PVC	66	37	1.4	0.9

Recycling: Ideals and Realities

Many materials COULD be recycled but very few ARE recycled



E-waste



10.3 million tonnes (Mt)
of waste generated in EU (2015)

40 %
is officially
collected

60 %
is lost



LOSSES OF E-WASTE



0.75 Mt
thrown in
waste bin



2,2 Mt
collected with
metal scrap



0,95 Mt
recycled under
non-compliant
conditions



0.75 Mt
scavenged
for valuable
parts



1,5 Mt
exported



E-waste contains precious metals and several critical raw materials such as gold, silver, antimony, beryllium, cobalt, neodymium and indium. A higher recycling rate of these materials would reduce their supply risk.



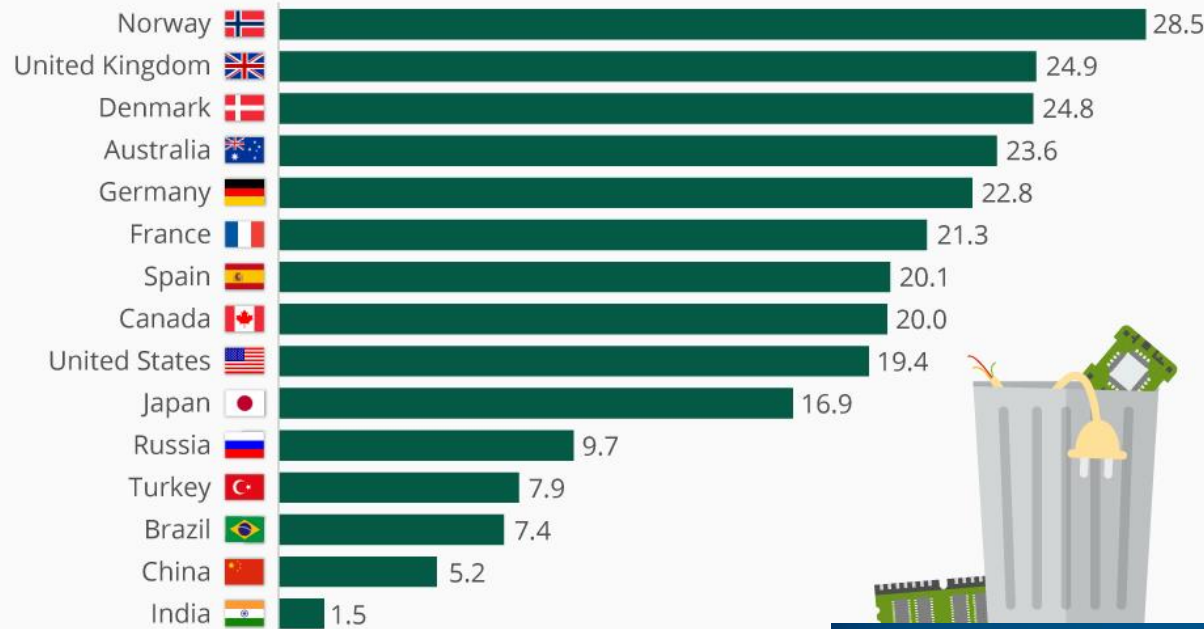
E-waste also contains several hazardous materials and chemicals such as halogenated compounds, radioactive substances, heavy metals and other metals that pose environmental and health risks if not managed adequately.

Source: EEA/ETC-WMGE

E-waste

The World's Worst Electronic Waste Offenders

E-waste generated per inhabitant in selected countries in 2016 (kg)



Electronic devices contain many small or micro – components and uses toxic or rare elements



Source: The Global E-waste Monitor, United Nations University via OECD
@StatistaCharts

Electronic waste is difficult to separate and recover rare earth elements

'Apple' are now collecting old iPhones just to recover the gold in them



Right to repair



E-waste is the fastest growing waste stream in the world
Only 15-20% is recycled



53 million tonnes of e-waste are produced each year the equivalent weight of 350 cruise ships' worth of electronics

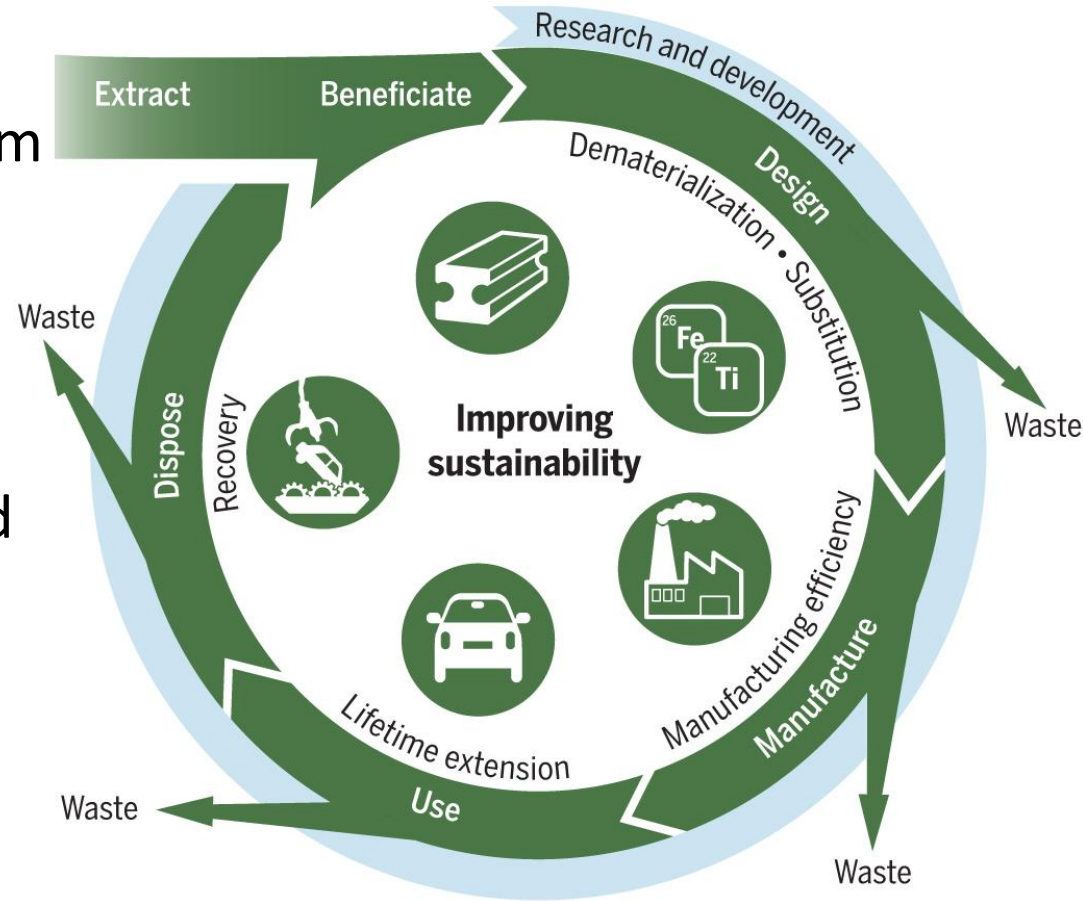
New '**right to repair**' laws in the UK and some USA States require electronic devices to be repairable and spare parts available for them



RIGHT TO
REPAIR

Design for Environmental Sustainability

- Design for environment means consider environmental impact from concept stages
- Selection of raw material based on sustainable resources
- Consider properties based on microstructure
- Include processing to improve sustainability
- Close the loop including upcycle and recycle



Summary



Materials selection and processing methods of materials determine their properties



Designing new materials or products requires a holistic approach that considers environmental sustainability from the conception to product use



LCAs are methods that can be used in the assessment of sustainability of material and product design, fabrication and future life